

WHAT NOISY FEEDBACK COSTS BAYESIAN OPTIMIZATION, AND THE ONE NUMBER THAT PREDICTS IT

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ABSTRACT

Bayesian optimization driven by human feedback is usually evaluated against regression oracles fitted to archival ratings. That entangles the cost of noisy feedback with the cost of the oracle being wrong, and a study of one design space cannot say what generalises. We instead inject four human-plausible error processes into twenty analytic landscapes with published optima, standardised so that a unit of error means the same on each: 76,800 paired noisy-versus-clean runs over four magnitudes, two onsets and twelve acquisition functions. Error as large as the landscape’s own standard deviation, present from the first rating, costs 14.1% of the available improvement; arriving at iteration 21 of 50, it costs 2.8%. In trials, one-SD error from the first rating means a median of 41 extra trials to match a clean 25-trial study within one percent of the achievable improvement, and 39% of runs never do within 100. At that magnitude the landscape and the onset move the cost by $8\times$ and $5\times$, the acquisition function and the kind of error by $2.8\times$ and $1.3\times$; over its range the magnitude itself moves it by $23\times$. Measured rater noise in three human studies falls where the loss is 12–20%. The expected loss from a single noisy greedy choice, computable without running an optimizer, explains nearly as much of the variation as the landscape descriptors and the error magnitude together, and stays informative beside a descriptor model that lets magnitude and landscape interact. Decomposing the deployed regret locates that cost: 61.8% [53.6, 69.5] of it is *selection* loss — the run visited a better design than the one it ships — the one term an acquisition function cannot reach, and identically zero under exact observation, so a noiseless benchmark never sees it. Consistently, none of seven acquisition-side changes improves the deployed design, and augmented expected improvement recovers 41% [27, 53] of the cost on the search trajectory and none of it on what ships. What works is spending trials on identification — a comparative sitting over the last third of the budget gains 0.051 [0.028, 0.072], and four confirming trials cut false improvement claims from 15.7% to 0.5% — or modelling the fault, worth 22–38% at near-zero price. Telling the surrogate the true noise variance removes only 5% of the cost, so the loss is informational; changing the incumbent definition removes 16% overall and up to 41% for one acquisition family, so robustness rankings rank acquisition-and-incumbent pairs. When the person applies the wrong design rather than the wrong rating, a slip of 15% of each coordinate’s range costs about as much as one-SD rating noise, a record of what was applied removes up to half of a small slip’s cost, and the design the log recommends does 1.6–2.6 $\times$ worse than the design the person experienced. Of the other process changes we tested, rating early designs twice, replication and qKG all do worse under error than the standard process, and re-rating the best designs before deploying helps only a little. On the same grid the fitted-oracle design reports about a third of the cost, and improving its oracles by up to 0.29 held-out R^2 does not close the gap.

1 INTRODUCTION

Bayesian optimization is the standard tool for optimizing an expensive black-box function from few evaluations (Jones et al., 1998; Shahriari et al., 2016; Frazier, 2018), and it is increasingly pointed at objectives that only a person can evaluate: interface parameters judged for trust or aesthetics, generated content ranked for quality, robot behaviours rated for comfort (Brochu et al., 2010; González et al., 2017). The optimizer assumes each observation is informative about the objective, but a person is not a sensor. People rate inconsistently, drift, are generous or harsh, make correlated errors, and sometimes apply the wrong setting altogether. How much that costs is asked routinely and, we argue, answered with a design that cannot support the answer.

The standard construction fits a regression model to archival ratings, treats it as the objective, injects synthetic error into its predictions, and measures the degradation. This has two problems that no care in the optimization code can fix. The fitted oracle is itself wrong — in the motivating studies its cross-validated R^2 as deployed is 0.55 at best and negative on one dataset — so the measured degradation mixes the injected error with the oracle’s mis-specification, and regret is measured against a random-search estimate of the optimum that the optimizer can legitimately beat. And the experiment runs on one design space, so a result such as “acquisition X is the most robust” has no way to say whether it holds anywhere else.

We remove both problems by giving up the human. On an analytic benchmark with a published optimum the objective is exact, regret is measured from the true optimum, and the geometry of the landscape can be measured before the optimization is run and used to predict what the error will cost. Twenty such landscapes turn “which acquisition is robust here” into “which property of a problem decides whether feedback error hurts”.

Contributions.

1. A protocol for measuring the cost of feedback error with no fitted oracle in the loop: twenty analytic landscapes with verified optima on a common scale, four human-plausible error processes, and a model-free control whose excess regret is exactly zero.
2. Measurements of that cost, in regret and in extra trials, and of what drives it (Sections 4–8). Magnitude, landscape and onset dominate, and a one-shot selection loss computable from the landscape and the noise level, without running an optimizer, predicts the cost.
3. A decomposition of deployed regret into search and selection loss (Section 5): selection is 61.8% of what error costs, is identically zero under exact observation, and is the term an acquisition cannot reach — which is what reporting regret at the best *observed* point hides.
4. An account of which remedies can therefore work (Section 9, Appendix N): over twenty-four budget-neutral arms, acquisition changes recover nothing on the deployed design, while spending a third of the budget on identification, or modelling the fault explicitly, recovers 22–38%, and four confirmation trials cut false improvement claims from 15.7% to 0.5%.
5. A matched run of the fitted-oracle design, which reports about a third of the cost, a gap that better oracles do not close (Appendix C).
6. An input-error arm, in which the person applies the wrong design rather than the wrong rating: a slip of 15% of the range costs about as much as one-SD rating noise, and up to half of a small slip’s cost is the record (Section 10). Six further checks, from the surrogate’s noise model to a preregistered replication on held-out seeds, each change one part of the design (Section 11).

2 RELATED WORK

Bayesian optimization under noise. Noise has been part of the formulation since Jones et al. (1998) and is routine in practice (Snoek et al., 2012), and several acquisitions are built for it: augmented expected improvement (Huang et al., 2006), the knowledge gradient (Frazier et al., 2009), and noisy expected improvement (Letham et al., 2019; Balandat et al., 2020). Picheny et al. (2013) benchmark infill criteria under noise on analytic problems, the closest antecedent to this work; we

differ in treating the error *process* and its onset as experimental factors, and in asking which landscape properties predict the outcome rather than which criterion wins. Heteroscedastic surrogates (Kersting et al., 2007) address a case we do not cover.

Human-in-the-loop and preference-based optimization. Where the objective is a person’s judgement, the usual response is to change the observation model: elicit pairwise preferences and fit an ordinal likelihood (Chu & Ghahramani, 2005; González et al., 2017). That handles inconsistent *scaling* but not inconsistent *ranking*, and it does not tell a practitioner running a cardinal-rating study what their rater noise will cost. Appendix O measures where the line falls: at equal human cost a comparison loop is *exactly* unharmed by drift — its shipped design differs from its clean twin in no run — and is no better under a saturating scale and worse under inconsistency. We trade the realism of the objective for exactness, and map measured rater noise onto the synthetic axis afterwards.

Benchmarks and landscape analysis. Results on analytic test functions need not transfer, and suites are small and idiosyncratic (Eggenberger et al., 2021). Here the twenty landscapes are observations in a regression whose unit is the landscape, and where that design cannot identify an effect we manipulate the geometry directly. Predicting algorithm behaviour from cheap landscape statistics is an old programme in evolutionary computation, from ruggedness (Weinberger, 1990) and fitness-distance correlation (Jones & Forrest, 1995) to their many descendants (Malan & Engelbrecht, 2013). Our predictor belongs to that family but is defined relative to the perturbation of interest, a property of the problem and the error level together.

3 SETUP

3.1 LANDSCAPES

We use twenty analytic benchmarks spanning $d = 2$ to $d = 11$: eight classical functions (Schwefel, Powell, Eggholder, Ackley, Shekel, Griewank, Hartmann-3, Hartmann-6), five further standard problems (Branin, Rosenbrock, Rastrigin, Michalewicz, Levy), six human-performance laws including the Yerkes–Dodson inverted-U (Yerkes & Dodson, 1908), and the moving-peaks benchmark (Branke, 1999). The suite was assembled for a separate study and adopted unchanged; Appendix H reports its geometry.

Each objective is affinely standardised to zero mean and unit variance over its own box, using a fixed scrambled-Sobol sample of 2^{16} points. Raw outputs span nearly seven orders of magnitude, so without this one grid of error magnitudes could not serve every landscape; the map is strictly increasing, so the argmax is unchanged. One scale is deliberately left alone. Write opt_z for the number of landscape standard deviations by which the optimum exceeds the mean, which is the expected regret of a random design. It ranges from 0.76 to 56.8, and that $74\times$ spread is what identifies which scale the cost of error follows (Section 6).

3.2 ERROR PROCESSES

Let $g(x)$ be the standardised objective, t the iteration, t_0 the onset and σ_e the swept magnitude, in landscape standard deviations. Observations are exact for $t \leq t_0$; thereafter the optimizer sees $\tilde{g}_t$:

$$\text{GAUSSIAN} \quad \tilde{g}_t = g(x_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (1)$$

$$\text{BIAS} \quad \tilde{g}_t = g(x_t) + \sigma_e + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (2)$$

$$\text{DRIFT} \quad \tilde{g}_t = g(x_t) + \sigma_e \frac{t-t_0}{T-t_0} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (3)$$

$$\text{AR}(1) \quad \tilde{g}_t = g(x_t) + e_t, \quad e_t = \rho e_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_e^2(1 - \rho^2)) \quad (4)$$

with $\rho = 0.8$ and e_{t_0+1} drawn from the stationary distribution, so the AR(1) process has standard deviation σ_e from its first step. BIAS models a systematically generous or harsh rater; its offset is scaled to the swept magnitude so that it and GAUSSIAN contrast a systematic with a random error of *the same size* (a fixed offset would be indistinguishable at the top of the grid). DRIFT models fatigue over a session, and AR(1) a rater whose consecutive judgements are correlated.

Noise first affects the observation at $t_0 + 1$; the first candidate that can react to it is proposed at $t_0 + 2$. We sweep $\sigma_e \in \{0.05, 0.25, 1, 5\}$ and $t_0 \in \{0, 20\}$ over $T = 50$ iterations with 5 initial uniformly

random samples, twelve acquisition functions and ten seeds: 79,200 runs, of which 76,800 form noisy/clean pairs.

3.3 OPTIMIZER AND ACQUISITIONS

Every arm uses the same surrogate, BoTorch’s default single-task GP (an RBF kernel with dimension-scaled lengthscales priors), inputs normalised to the unit cube and outputs standardised, fitted by marginal likelihood in BoTorch (Balandat et al., 2020). We run ten model-based arms — expected improvement and probability of improvement in their analytic, log (Ament et al., 2023) and Monte-Carlo forms (EI, LogEI, qEI, PI, LogPI, qPI), the upper confidence bound in analytic and Monte-Carlo form (Srinivas et al., 2010) (UCB, qUCB), noisy expected improvement (Letham et al., 2019) (qNEI), and pure exploitation of the posterior mean (Greedy) — and two model-free floors, uniform random sampling and a scrambled Sobol sequence. Improvement-based acquisitions take the maximum posterior mean at visited points as their incumbent, the choice usually recommended under noise and, as Appendix D.2 shows, not a neutral one.

3.4 METRICS

The primary response is post-onset per-iteration excess simple regret in landscape standard deviations: the regret of the noisy run minus that of the identically seeded clean run, averaged over the iterations the error could have affected. The two runs share their candidate-proposal random stream, so the difference isolates the error rather than the seed, and because y^* is a verified supremum rather than a random-search estimate, no regret can be negative. For reporting we divide by opt_z , the fraction of the achievable improvement destroyed; we never regress that fraction on opt_z or anything collinear with it, which would manufacture the coefficient Section 6 estimates.

3.5 CONTROLS

- **Model-free floor.** Random and Sobol never read an observation, so their excess regret must be identically zero, and across all 12,800 such comparisons it is exactly zero. This negative control shows that no observation noise reaches the candidate stream; the floors are then excluded from the robustness ranking, which a method that ignores its data would otherwise win.
- **Headroom.** Where the clean run already reaches the optimum a null is guaranteed, not earned; the smallest clean-run advantage over the floor is 0.176 (Rosenbrock), so no landscape is ceiling-limited.
- **Uniform offset.** A constant added to every observation is invisible to a GP that standardises its targets and uses a posterior-mean incumbent. With a shared noise stream the two arms propose identical designs to below 10^{-9} for the initial design and three model-based candidates (asserted by test), after which floating-point differences are amplified; in the sweep, BIAS and GAUSSIAN at onset 0 agree to within 0.012 at every magnitude.

4 HOW MUCH DOES FEEDBACK ERROR COST?

Table 1 gives the cost. Error whose standard deviation equals the landscape’s own, present from the first rating, costs 14.1% of the improvement that was available, and even error five times that large costs 22.0%. An optimizer that has already located a good region is hard to dislodge: the same 1σ error arriving at iteration 21 costs 2.8%, a fifth as much.

Table 1: Fraction of the achievable improvement destroyed by feedback error, averaged over landscapes and model-based acquisitions. Magnitudes are in landscape standard deviations.

error present	0.05σ	0.25σ	1σ	5σ
from it. 1	0.9%	5.3%	14.1%	22.0%
from it. 21	0.3%	1.2%	2.8%	4.9%

Table 2: Extra trials a noisy run needs to reach the regret its clean twin had after k trials, within 1% of the achievable improvement (budget-100 arm, GAUSSIAN error, six acquisitions): median over runs, and the share that never got there within 100 trials.

error	error from it. 1, match a clean 25-trial study	error from it. 41, match a clean 50-trial study
0.05σ	1 (4%)	0 (1%)
0.25σ	7 (14%)	0 (3%)
1σ	41 (39%)	0 (8%)
5σ	>78 (70%)	0 (12%)

Decomposing the variation by factor, at 1σ from the first rating unless stated, puts the practical conclusions in order. Sweeping the magnitude over its two decades moves the cost by $23\times$; changing the landscape moves it by $8.1\times$ (Powell 0.038 to Hartmann-6 0.303); moving the onset to iteration 21, by $5.0\times$; changing the acquisition function, by $2.8\times$ (PI to qUCB); and changing the error *process*, by $1.32\times$ (variance shares over all cells 0.24, 0.10, 0.17, 0.04, 0.002: onset and landscape trade places, the rest stands). The mean cost is 0.148 for GAUSSIAN, 0.152 for DRIFT, 0.147 for BIAS and 0.115 for AR(1): BIAS matches GAUSSIAN by construction at this onset (Section 3.5) and costs 20% more at the late onset, where the offset arrives as a step, and correlated error is gentler in every cell, partly absorbed into the surrogate’s mean.

Is 1σ a lot? We estimate within-rater noise in three published human studies [TODO: cite the three studies in the third person] with a nearest-neighbour (semivariogram nugget) estimator over repeated ratings by the same participant, and divide it by σ_f , the standard deviation of that study’s fitted objective over its design box (Table 20, Appendix H). Real rating noise lands between 0.74 and 3.35 landscape standard deviations, and participants are noisy from their first rating, so on the onset-0 row of Table 1 these studies would lose 12–20% of the achievable improvement, interpolating log-linearly between columns. The caveat is that σ_f can only be computed through a fitted oracle, and an under-fit oracle understates σ_f , so these ratios are if anything high; this anchors the range of interest rather than measuring it.

How many extra trials? A practitioner budgets trials, not regret. For each noisy run we count the trials it needs to come within 1% of the achievable improvement of its clean twin after k trials, on the budget-100 arm of Appendix F (GAUSSIAN error, six acquisitions; Table 2; all arms in Appendix J). Against a clean 25-trial study, error of 0.05σ from the first rating costs a median of one extra trial and 0.25σ seven; 1σ costs a median of 41, a budget of $2.6\times$, and 39% of runs never get there within 100 trials; at 5σ , 70% never do. Error from trial 41 costs a 50-trial study no extra trials at the median, though 8–12% of runs at 1σ and above never recover.

5 WHERE THE COST IS: SEARCHING OR SELECTING?

The regret reported above is the regret of the design a study would deploy, and it hides a distinction that decides which remedies can possibly work. Write x^* for the optimum, V for the designs a run visited and d for the one it deploys. Then

$$\underbrace{f(x^*) - f(d)}_{\text{deployed regret}} = \underbrace{f(x^*) - \max_{x \in V} f(x)}_{\text{search loss}} + \underbrace{\max_{x \in V} f(x) - f(d)}_{\text{selection loss}}. \quad (5)$$

The first term is what the optimizer failed to find. The second is what the ratings failed to identify among what it did find. An acquisition function acts on the first term only: it chooses where to sample, and the second term is already determined by the time the study picks a design to ship. The selection term is also exactly zero under exact observation — the best-rated visited design is then the best visited design — which is why it never appears in a noiseless benchmark and why the field’s habit of reporting simple regret at the best *observed* point is safe there and not here.

We evaluate both terms on all 64,000 noisy runs by re-scoring every visited design under the loop’s own refitted surrogate (Appendix K). As a check rather than an assumption, the largest selection loss over the 2,000 clean runs is 0.00×10^0 , as Equation 5 requires.

Table 3: Neither scale fits. Exponents of $\mathbb{E}[R] = A \sigma_e^{\beta_c} \text{opt}_z^{\beta_z}$ fitted to landscape $\times$ magnitude cell means, with 95% BCa intervals from 2000 landscape resamples and a jackknife over landscapes. β_z excludes zero in every condition; $\beta_c + \beta_z - 1$ excludes zero in six of eight.

error process	t_0	SPREAD: $\beta_z = 0$			GAIN: $\beta_c + \beta_z = 1$		verdict
		β_c	β_z	95% CI	$\beta_c + \beta_z - 1$	95% CI	
GAUSSIAN	0	0.51	0.86	[0.47, 1.13]	+0.37	[+0.15, +0.70]	neither
BIAS	0	0.51	0.85	[0.49, 1.08]	+0.36	[+0.17, +0.65]	neither
DRIFT	0	0.55	0.82	[0.55, 1.11]	+0.37	[−0.01, +0.63]	gain
AR(1)	0	0.66	0.85	[0.60, 1.15]	+0.51	[+0.14, +0.77]	neither
GAUSSIAN	20	0.62	1.15	[0.64, 1.48]	+0.76	[+0.03, +1.03]	neither
BIAS	20	0.70	1.14	[0.61, 1.58]	+0.85	[+0.04, +1.08]	neither
DRIFT	20	0.65	1.15	[0.61, 1.55]	+0.80	[−0.00, +1.14]	gain
AR(1)	20	0.71	1.14	[0.64, 1.49]	+0.85	[+0.13, +1.11]	neither

Most of what error costs is selection. Pooled over the four error processes, a deployed regret of 0.346 of the achievable improvement splits into 0.212 of search loss and 0.135 of selection loss, so selection is 38.9% [27.8, 52.6] of it. Against the identically seeded clean twin — the part the error caused rather than the part the problem is hard — of 0.218 of excess, 0.083 is search and 0.135 selection: **selection is 61.8% [53.6, 69.5] of the cost of error.** The share is the same for every error process (60.7–63.6%) and rises with magnitude and lateness of onset, from 29% at 0.05σ from the first rating to 82% at 5σ from trial 21.

No rule that re-reads the same ratings recovers it. Four other ship rules remove essentially none of the selection loss: +3.9% [−4, +11] for the posterior mean less two latent SDs, +0.3% for one, −1.7% for the mean rating, −8.7% for the posterior mean. The reason is structural: the posterior ranking the visited designs was fitted on the same corrupted ratings. Closing the term needs *new* information, which is what Section 9 spends trials on.

6 WHAT MAKES AN ERROR BIG?

Standardising every objective assumes that a landscape’s own standard deviation is the right unit for its error; the rival is the improvement available, since a half-SD error is one thing when the optimum stands 0.76 SDs above a random design and another at 56.8. We fit the power law

$$\mathbb{E}[R] = A \sigma_e^{\beta_c} \text{opt}_z^{\beta_z} \quad (6)$$

to the landscape-by-magnitude cell means of the raw excess regret R , using quasi-Poisson estimating equations, which keep the cells where noise helped (Appendix S). If the spread is the right unit (SPREAD), $\beta_z = 0$; if the error relative to the achievable gain is, $R = \text{opt}_z h(\sigma_e/\text{opt}_z)$, and for a power-law h that means $\beta_c + \beta_z = 1$ (GAIN). Intervals are BCa intervals over resampled landscapes, the unit of replication for a landscape-level covariate.

SPREAD fails at both onsets (Table 3): β_z excludes zero in all eight conditions, so standardising an objective does not make error magnitudes comparable, and a noise level reported without the problem’s headroom is under-specified. The cost also grows with opt_z faster than GAIN allows: $\beta_c + \beta_z - 1$ is +0.36 to +0.51 from the first observation and +0.76 to +0.85 from iteration 21, with intervals excluding zero in six of eight conditions, DRIFT touching it at both onsets; OLS on the logarithm of the positive cells rejects GAIN in all eight. The estimate leans on *shekel*, the one landscape with opt_z above 25; without it β_z is 1.04–1.10 and the gap 0.57–0.73 from the first observation. Appendix D.3 shows that opt_z is not what does the work: raising a narrow optimum changes nothing, and the dependence runs through how findable the optimum is.

7 A ONE-SHOT SELECTION LOSS PREDICTS THE COST

Both preceding sections describe the cost of error in terms of properties of the landscape, but neither gives a practitioner a number. Define $\text{frag}(\sigma_e)$ as the expected loss from a single greedy choice

Table 4: Acquisition robustness, ranked over landscapes within each condition and averaged over conditions. “Absolute loss” is the deployment number: the post-onset regret under noise as a percentage of the achievable improvement.

acquisition	mean rank	excess regret	absolute loss
qUCB	3.95	4.14%	27.9%
qNEI	4.34	4.89%	25.2%
UCB	4.41	4.73%	27.4%
LogEI	4.61	5.21%	26.0%
qEI	4.73	5.37%	25.5%
EI	4.98	5.42%	26.2%
qPI	6.29	7.90%	31.0%
Greedy	6.56	8.11%	32.5%
LogPI	7.48	9.19%	31.3%
PI	7.66	9.44%	31.7%
model-free floor	–	0.00%	41.0%

among m quasi-random candidates when each is observed with $\mathcal{N}(0, \sigma_e^2)$ error:

$$\text{frag}(\sigma_e) = \mathbb{E}_\epsilon \left[\max_i g_i - g_{\arg \max_i (g_i + \epsilon_i)} \right], \quad \epsilon_i \sim \mathcal{N}(0, \sigma_e^2), \quad (7)$$

where $g_i = g(x_i)$ over a fixed Sobol design. It involves no surrogate, no sequence and no acquisition function; at $m = 256$ it costs about 0.2 seconds per landscape.

With onset and error-process indicators in every model, $\text{frag}(\sigma_e)$ explains $R^2 = 0.52$ of the variation in excess regret across landscape-by-condition cells, nearly as much as the primary landscape descriptors and the error magnitude together (0.53), and it stays informative beside them ($\beta = +0.31$ [0.18, 0.45] in the interaction model) without making the magnitude redundant. Computed on a GP fitted to the first ten trials of a clean run rather than on the exact objective, it tracks the exact value at Spearman 0.94 and predicts the measured cost as well. Its residuals are positive in opt_z and tail weight: on tall, heavy-tailed landscapes fifty noisy choices cost more than one predicts (Appendix B).

8 WHICH ACQUISITION FUNCTION?

The confidence-bound family and noisy expected improvement are the most robust, and the probability-of-improvement family and pure exploitation (Greedy) fill the bottom four places on both robustness and absolute loss (Table 4). Averaged over conditions every model-based arm beats the model-free floor (41.0%); at 5σ from the first rating the floor matches or beats every acquisition on this metric, though not on the design it would ship. The Friedman test over the twenty landscapes as blocks reaches $p < 0.05$ in 28 of 32 conditions, and 115 of 288 pairwise comparisons against each condition’s empirically best arm survive within-condition Benjamini–Hochberg correction (Benjamini & Hochberg, 1995), a comparison that favours the winner.

The landscapes also disagree: the median Kendall’s W over the 32 conditions is 0.247, and qUCB takes 10 of 32 conditions, UCB and qNEI 6 each, LogEI 5, EI 3, qEI 2. A study run on one of these landscapes would report a defensible winner that largely does not transfer, and could not tell. Appendix D.2 shows that part of this ranking is not about the acquisitions at all.

A better search does not ship a better design. We ran seven acquisition-side changes as their own arms on the full suite, each keeping the human trials fixed, scored against the standard process (Appendix N). None improves the deployed design: augmented expected improvement (Huang et al., 2006) recovers +2% [−6, +11] of the cost of error, a lower-confidence-bound incumbent −27%, Thompson sampling −58% [−81, −40], a slip-aware acquisition −4% to −35%, a minimum-distance constraint −13%; redistributing the same rating effort across trials runs from −7% to −21%. Augmented EI makes the point alone: on the post-onset trajectory, the response an acquisition is designed to improve, it recovers +41% [+27, +53], the best of any arm we ran, and on

the design the study ships it recovers nothing. Under Equation 5 the term it improves is not the term that dominates.

9 BUYING IDENTIFICATION WITH THE TRIALS YOU HAVE

If most of the cost is selection, and no re-reading of the ratings recovers it (Section 5), then the only remaining lever is to spend human trials on identifying rather than searching. We hold the budget at $T = 50$ and spend the last k on one comparative sitting: the top k visited designs by the ship rule are each shown once, and the best-rated of them is deployed. Everything before trial $T - k$ is unchanged, so each procedure is replayed exactly from an existing run log (Appendix L). Error shared across the designs of one sitting cancels when they are compared, so only the idiosyncratic part enters the looks; we report both the full idiosyncratic SD and half of it.

Identification is worth about a third of the budget. Against the standard process, in units of the achievable improvement, a sitting of $k = 2$ gains $+0.017 [+0.004, +0.029]$, $k = 5$ $+0.035$, $k = 8$ $+0.042$, $k = 12$ $+0.046$, $k = 16$ $+0.051 [+0.028, +0.072]$, and then falls away: $k = 20$ $+0.042$, $k = 25$ $+0.022$, $k = 30$ -0.008 . The curve is the inverted U the two forces imply — a larger sitting is more likely to contain the best visited design and leaves fewer trials to find one — and it turns over at about a third of T . The ceiling, choosing the best k for each run in hindsight, is $+0.145$. Spending nothing extra and merely changing the ship rule gains $+0.001 [-0.018, +0.018]$, which is the selection result of Section 5 in another currency: the trials, not the rule, are what buys the recovery.

Ship the fresh look, not the model. Of the 19 variants replayed, the best takes the top five candidates by the cautious ship rule and deploys whichever the final sitting rates highest: it recovers 19.7% [13, 27] of the deployed cost of GAUSSIAN error, 33.6% [25, 41] at 5σ from trial 21, and 27.9% [20, 36] under an unnoticed slip. Folding the same looks into the surrogate’s posterior rather than trusting them does worse (11.5%), for the reason the selection term exists at all: the posterior was fitted on the corrupted ratings and carries them forward. It is not universally safe — under misclicks it *loses* 14.3% $[-30, -2]$, since a sitting re-shows the *recorded* design rather than the one that earned the rating, and below 0.25σ every variant costs more than the error does.

A model-based budget rule inherits the corruption it was meant to fix. Choosing k per run from the surrogate’s posterior and the run’s own rate of improvement (Appendix M) *loses* to a fixed k : $+0.004 [-0.013, +0.021]$ against $+0.051$, because in 65% of runs it declines the sitting for the posterior-mean design, the worst policy of all at -0.013 . Its utility is the surrogate’s posterior, which was fitted on corrupted ratings, so it is confident about designs it should not be. Nothing we tried beat fixing k at about a third of the budget.

10 WHEN THE ERROR IS IN THE DESIGN, NOT THE RATING

Every error process above corrupts the reported value. People also make the other mistake: a slider overshoots, the wrong option is pressed. The optimizer proposes x , the person experiences $x' \neq x$, rates it honestly, and the rating is filed against x — the right value at the wrong location, which a fitted oracle cannot score at all, since scoring it needs the objective at x' . We sweep a SLIP (every trial displaced by $\mathcal{N}(0, s^2)$ per coordinate) and a MISCLICK (a uniformly random design with probability p); two results carry (Appendix A). A slip of 15% of each coordinate’s range costs about what one-SD rating noise costs, so this is not a second-order concern. And **the design in the log is not the design that earned its rating**: scored at the recorded x a practitioner would deploy, the final design’s excess regret is 16.4% of the achievable improvement at a 5% slip and 65.8% at 40%, against 6.5% and 24.9% for the design the person actually experienced — a factor of 2.5 to 2.6, of which a record of what was applied removes up to half. The optimizer’s progress survives an input error far better than the record of it, and a deployment inherits the record.

11 THE CONCLUSIONS UNDER SIX FURTHER CHECKS

The surrogate’s noise estimate is not the problem. The GP’s fitted noise is an eighth of the injected value for the first ten iterations at 1σ and above, yet telling it the true variance removes

5.0% of the excess-weighted cost, 3% or less at 1σ and above, and no condition survives correction (Appendix D.1).

The incumbent is part of the method. Switching improvement-based acquisitions from the posterior-mean incumbent to the observed maximum removes 15.6% of the excess-weighted cost (35–41% for the probability-of-improvement family) but worsens the deployed design: robustness rankings rank acquisition-and-incumbent pairs (Appendices D.2 and Q).

opt_z is not causal on its own, and dimension is not inert. Purpose-built landscapes show that raising a narrow optimum changes nothing, that a Levy ladder confounds dimension with signal strength, and that with opt_z fixed dimension still raises the cost, $1.7\times$ from $d = 7$ to 11 (Appendix D.3).

Multi-objective BO is no more fragile. On six BoTorch problems with published hypervolume optima the dose-response and onset structures replicate, early error costs at most half as much as on the scalar suite, and the four hypervolume acquisitions are indistinguishable (Appendix E).

The budget matters; the rating scale cuts both ways. With the late onset held at 40% of the run, the early-to-late cost ratio is $2.3\times$, $5.1\times$ and $11.7\times$ at $T = 25, 50$ and 100 ; a discretised rating scale makes small errors costlier and large ones marginally cheaper; and scored against the standard process, neither qKG nor replication recovers the cost (Appendices F and Q).

The replication is void; post hoc, the claims hold. Four claims were preregistered before ten fresh seeds ran; one control failed (a landscape fell below the headroom threshold), so the run is void by its own rule, and with that landscape excluded post hoc all four claims clear their thresholds (Appendix G).

The fitted-oracle design reports about a third of the cost. Run on three archival datasets at matched magnitudes, the data-driven design recovers 29–38% of the cost the analytic arm measures. The gap is not the oracles’ fidelity — improving them by up to 0.29 held-out R^2 moves the measured cost by a median of $+0.006$ — nor the optimum estimate, which cancels in a difference of two regrets; the three fitted design spaces simply sit at the low end of the twenty-landscape spread (Appendix C).

12 DISCUSSION

At measured rater-noise levels, feedback error costs a study 12–20% of the improvement it could have had: worth reducing, but not a catastrophe, and not one extra trials repair, since 39% of runs at 1σ never match a clean study with four times its budget. Classifying the kind of error is largely wasted effort: at 1σ and above the four processes are within a third of each other. Estimating its size is not: computed from a ten-trial pilot and the rater noise, Equation 7 predicts the damage as well as the exact version does. When the error is in the design, recording what was applied removes up to half of a small slip’s cost, and the design a log recommends should be re-evaluated before it ships.

Most process changes do not help either (Appendix Q). Replicating ratings, early or throughout, does worse under error than the standard process: in fifty trials the designs it gives up cost more than the averaging buys. qKG loses on the deployed design. The observed-max incumbent recovers a fifth of the trajectory cost but deploys a worse design, and re-rating the top three designs before deploying recovers 4% of the deployed design’s cost.

Two things remain unexplained: why the onset effect grows with the budget (Appendix F), and why the cost grows with opt_z faster than GAIN allows when opt_z is not causal (Appendix D.3).

Limitations. The landscapes are analytic, not people: they remove the surrogate-human confound at the cost of realism, and Appendix C cannot say which design’s answer is closer to a person. The rater-noise anchor runs through a fitted oracle’s σ_f ; extra-trial counts are censored at the budget and depend on the tolerance (Appendix J). The multi-objective arm is six problems at five seeds under GAUSSIAN error, too few to separate the hypervolume acquisitions. The preregistered replication

is void on a control failure, so the claims clear their thresholds only after a post-hoc exclusion and cannot be called confirmed.

REPRODUCIBILITY STATEMENT

The full pipeline is scripted end to end, from the landscape statistics through the sweep to every table in this paper, and every stochastic component is explicitly seeded. Regenerating published LogEI and qNEI runs from their recorded settings reproduces them bit for bit, and the seed order of the model-based acquisitions has not changed since the sweep. The model-free floors of the main sweep do not reproduce their recorded noisy observations: their noise seed moved when two robust baselines were added to the acquisition list after that sweep ran. Their candidates and regret still reproduce exactly, since a floor never reads an observation. No reported number depends on those, and the seed order is now pinned by test. Three further checks are automated. The vendored benchmark implementations are asserted by test to agree with their source implementations to 10^{-9} relative on random points, so the objective cannot silently drift. The model-free control of Section 3.5 is verified to be exactly zero in every run. And one recorded optimum in the source suite, the supremum for a superseded parameter setting, is wrong by 64%; we detected this by searching every box against its recorded value and corrected it here; the correction and the provenance of the original are both asserted by test. [TODO: repository URL and commit at camera-ready.]

LLM USAGE

A large language model (Claude, from Anthropic) was used extensively in this work. It proposed and implemented parts of the experimental design and analysis, wrote and tested most of the simulation and analysis code, ran and monitored the experiments, drafted and edited the manuscript, and checked the numbers in the text against the analysis outputs. The authors set the research questions, chose among the proposed designs, and are responsible for everything reported here. [TODO: authors: confirm or amend this description before submission.]

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Table 5: Fraction of the achievable improvement destroyed by an input error. A slip’s magnitude is the displacement SD as a fraction of each coordinate’s range; a misclick’s is the probability of a random trial. The last three columns decompose the early-onset slip: *floor*, the model-free arms; *logged*, a learner whose record holds the point evaluated; *mislabelling*, the paired difference between the unnoticed slip (second column) and *logged*. Brackets: 95% intervals over landscapes.

magnitude	SLIP		MISCLICK		SLIP from it. 1, decomposed		
	from it. 1	from it. 21	from it. 1	from it. 21	floor	logged	mislabelling
1%	1.5 [0, 3]	1.1 [0, 2]	0.4 [0, 1]	0.1 [0, 0]	0.3 [0, 1]	0.7 [0, 2]	0.8 [0, 2]
5%	5.4 [3, 8]	3.0 [2, 5]	2.8 [2, 4]	0.5 [0, 1]	0.2 [-1, 1]	2.8 [2, 4]	2.7 [1, 4]
15%	11.5 [8, 15]	5.0 [3, 8]	6.8 [5, 9]	2.0 [1, 3]	1.2 [-1, 3]	7.1 [5, 10]	4.4 [2, 6]
40%	23.2 [16, 31]	6.4 [4, 10]	13.5 [10, 18]	3.5 [2, 6]	6.4 [0, 13]	19.6 [12, 28]	3.6 [2, 6]

A THE INPUT-ERROR ARM

Every error process above corrupts the reported value. People also make the other mistake: a slider overshoots, the wrong option is pressed, a configuration is applied to the wrong condition. The optimizer proposes x , the person experiences $x' \neq x$, rates it honestly, and the rating is filed against x : the right value at the wrong location. A fitted oracle cannot score it, since that needs the objective at x' .

A SLIP displaces every trial by $\delta \sim \mathcal{N}(0, s^2)$ per coordinate, with s a fraction of that coordinate’s range, and clips the result to the box. A MISCLICK leaves the trial exact with probability $1 - p$ and otherwise lands on a uniformly random design. Both magnitudes are swept over $\{0.01, 0.05, 0.15, 0.4\}$ at the same onsets and budget, with six model-based acquisitions (LogEI, EI, PI, UCB, qUCB, qNEI), the two floors and five seeds; neither is in landscape standard deviations, so the comparison with the response-error grid is on the damage axis only.

If the slip goes unnoticed the surrogate trains on $(x, g(x'))$ and the deployed design is the recorded x , so we score it there. A counterfactual arm records $(x', g(x'))$, as an instrumented interface would, with identical slip draws; the contrast separates the cost of evaluating the wrong point from that of mislabelling it. The model-free floors are no longer a zero control: they evaluate the wrong points without learning, so their excess is the geometric cost of a slip alone.

A slip costs about as much as a rating error. A slip of 5% of each coordinate’s range on every trial, from the first, destroys 5.4% of the achievable improvement; one of 15% destroys 11.5%; one of 40%, 23.2% (Table 5). On this arm’s six acquisitions and seeds under GAUSSIAN error, the 0.25σ , 1σ and 5σ rating errors of Table 1 cost 3.8%, 11.7% and 21.3% (the $T = 50$ row of Table 15; pooled: 5.3%, 14.1%, 22.0%): a 5% slip costs more than a 0.25σ rating error, a 15% slip about as much as 1σ , a 40% slip about as much as 5σ . A misclick is cheaper at every matched magnitude: even when 40% of trials land somewhere random it costs 13.5%, little more than a 15% slip on every trial and barely half of a 40% one. A late slip is cheaper than an early one by less than a late rating error: $1.4\text{--}3.6\times$ against $5.1\times$ at 1σ on the same basis.

Half of a small slip’s cost is the record, not the displacement. The decomposition in Table 5 (in full, Table 22) separates the mechanisms. The model-free floor barely moves, since a random design displaced is still random; only at 40%, where clipping piles evaluations at the box edges, does it lose 6.4% [0, 13]. A learner with a correct record loses far more, 2.8% at a 5% slip and 19.6% at 40%: a displaced proposal is a worse one. Mislabelling adds 2.7 points at 5% and 4.4 at 15%, half and two-fifths of the total (paired intervals [0.35, 0.61] and [0.26, 0.49]; not estimable at 1%); at 40% the displacement dominates and the mislabelled share falls to a sixth.

The design in the log is not the design that earned its rating. Scored at the recorded x a practitioner would deploy, the final design’s excess regret is 16.4% of the achievable improvement at a 5% slip from the first trial, 34.3% at 15% and 65.8% at 40%, against 6.5%, 13.4% and 24.9% for the best design the person experienced (Appendix I): a factor of 2.5 to 2.6, and 1.6 at a 1% slip. The optimizer’s progress survives an input error far better than the record of it, which is what a deployment inherits.

Table 6: Predicting the cost from one noisy choice. Fitted on landscape $\times$ condition cell means, predictors z -scored, every model also carrying onset and error-process indicators; the interaction models add $(\log \sigma_e)^2$ as well. Intervals and p -values from a 2000-replicate bootstrap over landscapes.

model	frag(σ_e)	$\log \sigma_e$	$\log \text{opt}_z$	$\log \sigma_e \times \log \text{opt}_z$	$\log \text{tail}$	rugged.	d	R^2
frag only	+0.75***	—	—	—	—	—	—	0.520
descriptors only	—	+0.37***	+0.64***	—	+0.20**	−0.09	−0.03	0.526
both	+0.66***	−0.07	+0.42**	—	+0.20**	−0.06	−0.03	0.701
descriptors + int.	—	−0.25	+0.86***	+0.90***	+0.20**	−0.09	−0.03	0.779
both + int.	+0.31***	−0.38**	+0.70***	+0.67***	+0.20**	−0.08	−0.03	0.792

B PREDICTING THE COST FROM ONE NOISY CHOICE

The regression behind Section 7, whose predictor is Equation 7.

With onset and error-process indicators in every model, $\text{frag}(\sigma_e)$ explains $R^2 = 0.52$ of the variation in excess regret across landscape-by-condition cells (0.48 on its own), nearly as much as the primary landscape descriptors together with the error magnitude ($R^2 = 0.53$; Table 6). Jointly they reach $R^2 = 0.70$, and in that additive model the error magnitude collapses to $\beta = -0.07$ with $p = 0.44$. That null belongs to the additive specification: with the magnitude-by- opt_z interaction Section 6 found, and a quadratic in the magnitude, the descriptor model reaches $R^2 = 0.78$ with no frag term; frag still adds to it, keeping $\beta = +0.31$ [0.18, 0.45] at $R^2 = 0.79$, but the magnitude is significant again beside it (−0.38 [−0.60, −0.08]). **One noisy choice predicts, on its own, two thirds of what the best descriptor model explains, and stays informative beside it; it does not make the error magnitude redundant.** Dimension and ruggedness are not detectably different from zero here (intervals [−0.26, 0.12] and [−0.34, 0.15]), though the fixed- opt_z ladder of Appendix D.3 shows dimension does raise the cost.

Equation 7 needs the objective at a quasi-random design (here 256 points) and an estimate of rater noise; computed instead on a GP fitted to the first ten trials of a clean run, it tracks the exact value (Spearman 0.94) and predicts the measured cost as well (0.85 against 0.84; Appendix R). Its residuals are positive in opt_z (+0.42) and tail weight (+0.20): on tall, heavy-tailed landscapes fifty noisy choices cost more than one predicts.

C WHAT THE FITTED-ORACLE DESIGN WOULD HAVE CONCLUDED

The introduction argued that fitting a regression oracle to archival ratings entangles the cost of feedback error with the cost of the oracle being wrong. To measure this rather than argue it, we run the data-driven simulator on the three archival datasets at the same error magnitudes, expressed in each dataset’s own σ_f so that “1 σ ” means the same thing in both arms.

Table 7: The same experiment, run against exact functions and against oracles fitted to ratings, under GAUSSIAN error. Percentages are post-onset excess as a fraction of the gap between the optimum and a same-budget model-free floor (the floor gap), the one normaliser computable in both arms. The fitted design reports roughly a third of the cost.

arm	error from it. 1				error from it. 21			
	0.05 σ	0.25 σ	1 σ	5 σ	0.05 σ	0.25 σ	1 σ	5 σ
known functions (20)	10	44	99	148	3	7	10	16
fitted oracle (3)	1	11	33	47	1	3	6	8
fitted, no augmentation (3)	2	14	38	50	1	4	6	7

The fitted arm has no verified optimum, so Table 7’s denominator uses the best value any arm reached, which is attainable by construction and so inflates the fitted fraction: the bias works against the result below. Three design spaces against twenty also widen the intervals.

The fitted design understates the cost of feedback error by roughly threefold. At 1σ from the first rating, the known-function arm puts the loss at 99% of the floor gap (cluster-bootstrap interval [55, 155]) and the fitted arm at 33% ([5, 75]): a ratio of 3.0, interval [1.1, 17]. At 0.05σ the fitted arm reports 1% ([−4, 3]), indistinguishable from no effect, where the exact functions report 10% ([4, 18]). The direction holds in every cell (twofold at the late onset above 0.05σ), though the intervals overlap at the larger magnitudes.

Oracle fidelity does not explain it. Turning off the deployed jitter augmentation improves the companion’s oracles by up to 0.29 held-out R^2 (opticarvis, 0.52 to 0.81). Rerunning the companion arm without it shifts the fraction destroyed by +0.016 (median +0.006) over the 2,400 shared cells, and the augmentation-free arm is the worse of the two in 52.6% of them. If the gap were fidelity, it would close as the oracle improved. Nor can the optimum estimate be the cause: it cancels in the excess, a difference of two regrets, and enters only the denominator, which biases the other way. The three fitted design spaces sit at the low end of the twenty-landscape spread (per-landscape fractions at 1σ run from 0.11 to 4.8; the fitted three are 0.05, 0.19 and 0.75), and with three design spaces the arm cannot say whether that is a property of human objectives or of these studies.

D ABLATIONS

D.1 DOES THE SURROGATE KNOW IT IS BEING LIED TO?

The GP fits its observation noise as a free hyperparameter under a prior that shrinks it, so a measured cost of noise could in principle be the surrogate’s failure to notice the error rather than the information the error destroys. This is the first thing to rule out, because if it were the explanation then every number above would be a statement about BoTorch’s default prior rather than about noisy feedback.

The mis-estimation is real, and it lands where it would do the most damage (Table 8). Refitting each surrogate from the sweep’s own logs using only its first n observations, at 1σ and above the fitted noise standard deviation is about an eighth of the injected one for the first ten of fifty iterations (fifteen observations) and has largely recovered by iteration twenty, while at 0.05σ it is overestimated by a quarter to a half throughout — so a diagnostic evaluated only at $n = 50$ looks reassuring and is misleading. Refitting with a wide noise prior recovers the injected level far sooner (0.55 rather than 0.14 at $n = 15, 1\sigma$), so the shrinkage is the prior’s doing and not a limit of the data.

Table 8: Fitted over injected observation-noise standard deviation, as a function of how many observations the surrogate has seen, for error present from iteration 1. The estimate is badly wrong exactly while the trajectory is being decided — and Table 9 shows that this costs almost nothing.

observations	0.05σ	0.25σ	1σ	5σ
8	1.43	0.30	0.12	0.11
15	1.50	0.41	0.14	0.10
25	1.37	0.72	0.66	0.56
50	1.24	0.88	0.83	0.89

It was natural to expect this to explain the onset effect of Section 4: from the first observation the error arrives in exactly the window where the noise cannot yet be estimated, whereas at a mid-run onset the surrogate has a settled length scale and thirty iterations in which to learn. **We tested that directly and it is wrong.** Supplying the true injected variance as `train.Yvar` and re-running 4,800 paired cells removes only 5.0% of the excess-weighted cost (40% at 0.25σ , not significant; 3% or less at 1σ and above), no condition survives FDR correction (Table 9), and the onset-0 to onset-20 ratio at 1σ does not shrink — it goes from 4.0 to 5.2. Per landscape the largest reduction is Ackley at 47%; per acquisition the effect is inconsistent in sign, helping PI (22%) and qNEI (15%) but *hurting* qUCB (−29%).

Two conclusions follow. Bayesian optimization is remarkably insensitive to its own noise hyperparameter in this regime, despite getting it badly wrong for the first fifth of the budget — worth knowing on its own, and a caution against reading too much into noise-estimation diagnostics. And the cost this paper measures is the information cost of the error, not an artefact of a mis-specified

surrogate, which is what licenses reading Table 1 as a property of the problem. The onset effect itself is therefore left unexplained; we return to it in Section 12.

Table 9: Telling the surrogate the truth about its noise. Paired within (landscape, acquisition, magnitude, onset, seed); positive Δ means the treatment suffers more. Nothing survives FDR correction.

σ_e	t_0	reference	treatment	Δ	share	d_z	p_{FDR}
0.05	0	0.065	0.077	+0.012	−19%	0.03	0.888
0.25	0	0.271	0.166	−0.105	+39%	−0.18	0.888
1	0	0.790	0.848	+0.058	−7%	0.10	0.888
5	0	1.484	1.463	−0.021	+1%	−0.04	0.888
0.05	20	0.002	0.001	−0.002	+71%	−0.02	0.913
0.25	20	0.072	0.041	−0.031	+43%	−0.20	0.888
1	20	0.197	0.164	−0.034	+17%	−0.19	0.888
5	20	0.531	0.483	−0.048	+9%	−0.16	0.888

D.2 IS IT THE ACQUISITION OR THE INCUMBENT?

best_f is the maximum posterior mean at visited points, which shrinks under noise and so makes improvement-based acquisitions automatically more exploratory. The acquisitions differ in their exposure to this: LogEI, EI, PI, LogPI and qEI consume best_f ; UCB does not; qNEI ignores it entirely. A ranking of acquisitions by robustness is therefore partly a ranking of their sensitivity to a design choice that is not part of the acquisition at all. Re-running with the observed maximum as the incumbent removes 15.6% of the excess-weighted cost overall (14% at 1σ , two thirds or more at 0.25σ and below), with four of eight conditions significant after FDR correction (Table 10).

The per-acquisition breakdown is in Table 11. UCB and qNEI move by *exactly* zero, as they must — a control that validates the pairing at machine precision. The probability-of-improvement family moves most: LogPI loses 41% of its measured cost and PI 35%, against 17% for EI, 2% for LogEI and −2% for qEI. So a substantial part of “PI is fragile under noise” is really “the posterior-mean incumbent is fragile under noise, and PI is the acquisition most exposed to it”. Every acquisition-robustness ranking here, including Table 4, is a ranking of acquisition-and-incumbent pairs, and the same is true of the applied results it is modelled on. Reporting the incumbent definition alongside the acquisition should be standard. Scored against the standard process’s clean run rather than its own, the switch still recovers a fifth of the trajectory cost, but the design it deploys is worse (Appendix Q).

Table 10: Observed-maximum against posterior-mean incumbent, same pairing as Table 9.

σ_e	t_0	reference	treatment	Δ	share	d_z	p_{FDR}
0.05	0	0.082	0.027	−0.054	+67%	−0.54	0.046
0.25	0	0.308	0.109	−0.199	+65%	−0.46	<0.001
1	0	0.888	0.781	−0.107	+12%	−0.50	0.048
5	0	1.478	1.458	−0.020	+1%	−0.09	0.085
0.05	20	0.017	−0.013	−0.030	+179%	−0.27	0.117
0.25	20	0.119	−0.001	−0.120	+101%	−0.41	0.001
1	20	0.242	0.189	−0.053	+22%	−0.26	0.053
5	20	0.581	0.586	+0.005	−1%	0.02	0.573

Table 11: The incumbent arm by acquisition. UCB and qNEI never read best_f and move by exactly zero, which validates the pairing; the probability-of-improvement family moves most.

acquisition	reference	treatment	Δ	share removed
LogPI	0.555	0.328	-0.227	+41%
PI	0.572	0.373	-0.199	+35%
EI	0.459	0.382	-0.077	+17%
LogEI	0.493	0.481	-0.011	+2%
qNEI	0.378	0.378	+0.000	+0%
UCB	0.346	0.346	+0.000	+0%
qEI	0.445	0.454	+0.009	-2%

D.3 BREAKING THE DESCRIPTOR CONFOUND

In our suite opt_z , sparsity and skew correlate at about 0.9, and for a bounded function with an isolated optimum that is close to a structural identity, so Section 7’s descriptor terms cannot be separated observationally, however many real benchmarks are added. Two purpose-built families vary them by construction: a cos-field background plus one spike, whose amplitude moves opt_z and whose width moves sparsity; and the Levy function at $d \in \{4, 7, 11\}$, one landscape family across a dimension ladder (Table 12).

opt_z on its own is not causal. Holding the spike narrow and raising its amplitude by $4\times$ moves opt_z from 5.0 to 21.7 and changes the excess regret at 1σ from 0.246 to 0.245 — no effect at all. Widen the spike and the same amplitude change raises excess from 0.495 to 1.888. What the observational $\beta_{\text{opt}_z} = +0.42$ picks up is therefore an interaction, not a main effect: a taller optimum costs more to miss only when it is broad enough that the optimizer would otherwise have found it. On a true needle, the optimizer was never going to find the optimum with or without noise, so there is nothing for the noise to destroy. It also means that the opt_z exponent of Section 6 describes the suite rather than a mechanism: it is not opt_z as such that makes an error expensive.

Dimension, and a within-family ladder that corrects itself. Within the Levy family, excess regret at 1σ rises monotonically from 0.114 at $d = 4$ to 0.222 at $d = 7$ and 0.264 at $d = 11$, while the cross-suite regression puts β_d at -0.03 and not significant. Read on its own that looks like a real effect the observational analysis missed. It is not, or not mostly: opt_z also rises along the Levy ladder, 1.53 to 2.07 to 2.62, so that comparison varies dimension and signal strength together.

The bump ladder separates them. Amplitude and width are chosen at each dimension so that opt_z is 9.0 and the spike occupies a 10^{-6} fraction of the box at all three, leaving dimension as the only difference. In the table’s units, excess regret is 0.28 at $d = 4$, 0.25 at $d = 7$ and 0.43 at $d = 11$. The $d = 4$ rung has headroom 0.098, just below the 0.10 screen applied everywhere else, so the comparison that clears the screen is $d = 7$ to 11: there the fixed- opt_z ladder rises $1.7\times$ where Levy rises $1.2\times$. **Dimension does raise the cost when opt_z is held fixed; what the Levy ladder cannot say is how much of its $2.3\times$ is dimension rather than signal strength.** What rises monotonically with dimension on both ladders is the inference metric, the true value of the design a practitioner would deploy: 0.52, 0.70, 0.99 on the fixed- opt_z ladder. Higher dimension makes the progress harder to identify from noisy observations at the end as well as harder to make.

The methodological lesson is that a within-family ladder is not automatically a manipulation of the property it is indexed by: the Levy ladder varied signal strength along with dimension, and ours became a manipulation of dimension only when it was built to hold opt_z fixed.

Table 12: Purpose-built landscapes. Raw post-onset excess regret in landscape standard deviations, GAUSSIAN error from iteration 1. Amplitude and width are varied independently in the top block; the middle block is one function family at three dimensions; the bottom block holds $\text{opt}_z = 9.0$ and the spike volume fixed across three dimensions.

landscape	d	opt_z	sparsity	0.05σ	0.25σ	1σ	5σ
bump_a4_w0.05	4	5.0	$<1.5\text{e-}05$	0.007	0.152	0.246	0.324
bump_a16_w0.05	4	21.7	$<1.5\text{e-}05$	0.014	0.026	0.245	0.262
bump_a4_w0.15	4	5.3	$3.8\text{e-}04$	0.133	0.289	0.495	0.704
bump_a16_w0.15	4	12.3	$1.8\text{e-}04$	0.298	0.730	1.888	3.173
levy_4d	4	1.5	$2.6\text{e-}01$	0.006	0.048	0.114	0.200
levy_7d	7	2.1	$7.3\text{e-}02$	0.051	0.115	0.222	0.305
levy_10	11	2.6	$1.3\text{e-}02$	0.001	0.147	0.264	0.481
bump_d4	4	9.0	$<1.5\text{e-}05$	0.001	0.121	0.279	0.327
bump_d7	7	9.0	$<1.5\text{e-}05$	-0.001	0.103	0.248	0.428
bump_d11	11	9.0	$<1.5\text{e-}05$	0.142	0.215	0.427	0.734

E DOES ANY OF THIS HOLD FOR MULTIPLE OBJECTIVES?

The applied setting is frequently multi-objective (Daulton et al., 2021) and everything above is scalar, so the generalisation cannot be assumed. We repeat the design on seven BoTorch problems with a published maximum hypervolume — BraninCurrin, ZDT1, DTLZ2, DH2, VehicleSafety, Penicillin and CarSideImpact — spanning two to four objectives and two to seven dimensions, with the four hypervolume acquisitions (qEHVI, qNEHVI and their log variants) and the same two model-free floors, under GAUSSIAN error, at five seeds: 1,680 paired runs, of which 960 are model-based noisy-versus-clean comparisons on the problems that survive the headroom screen below.

Standardisation transfers exactly rather than approximately. Each objective is mapped to zero mean and unit variance over a fixed scrambled-Sobol sample of 2^{15} points, drawn as for the scalar suite, and under a per-objective affine map the hypervolume scales by $\prod_m \sigma_m$, so the standardised maximum follows in closed form from the published one. We verify this rather than assert it: across the seven problems the largest relative error between the closed-form standardised maximum and a direct recomputation is 4.7×10^{-16} .

Controls. The model-free floor is again identically zero across all 560 such comparisons, so no observation noise leaks into the candidate stream in the hypervolume code path either. No run beats its published maximum hypervolume. Acquisition-optimization fallbacks total zero, which we report rather than assume: a hypervolume acquisition that fails to construct falls back to random sampling, and because that produces a plausible-looking curve rather than an error, the arm has to be read off the fallback counter and not off the plots.

One problem is dropped, by the same screen as before. DH2’s reference point leaves a 32,768-point Sobol sample with a hypervolume of exactly zero, and its best learner does not improve on the model-free floor at all. There is nothing for error to destroy, so its null is guaranteed by construction; it is excluded. The remaining six pass, with Penicillin weakest at 0.197 — comparable to Rosenbrock at 0.176, the weakest admissible scalar landscape.

The comparison needs one piece of care. Elsewhere the paper divides excess regret by opt_z , a z-score of the optimum against a random design’s mean. A Pareto problem has no scalar optimum and therefore no opt_z , and the natural hypervolume analogue — the gap between the published maximum and what a model-free design of the same budget reaches — is a numerically much smaller denominator. Quoting one arm’s published fraction against the other’s would manufacture a threefold difference out of the choice of divisor alone. Both columns of Table 13 are therefore recomputed on the floor denominator, identically; and we also report the onset ratio, which is a ratio of two costs on the same problem and so does not depend on the denominator at all.

Table 13: The two arms on a common footing. Both columns are post-onset excess divided by the gap between the optimum and a same-budget model-free floor, as a percentage; brackets are 95% cluster-bootstrap intervals over landscapes/problems. This is *not* the normalisation used in Table 1, and the two must not be read against each other; the scalar column is recomputed from the same runs as the known-function row of Table 7, under GAUSSIAN error.

error	scalar (20 landscapes)		multi-objective (6 problems)	
	from it. 1	from it. 21	from it. 1	from it. 21
0.05σ	10.2 [4, 19]	2.8 [1, 5]	1.4 [0, 3]	0.5 [-1, 2]
0.25σ	44.3 [22, 71]	6.5 [4, 10]	13.6 [8, 19]	3.7 [0, 10]
1σ	99.0 [54, 154]	10.3 [7, 13]	49.1 [42, 56]	10.7 [2, 24]
5σ	148.4 [76, 239]	16.5 [11, 22]	72.3 [55, 91]	13.0 [3, 27]

The two headline structures replicate. Cost rises monotonically with error magnitude and falls sharply when the error arrives late, on both arms. At 1σ from the first rating the multi-objective arm loses 49.1% of the hypervolume floor gap [42, 56] against the scalar arm’s 99.0% [54, 154]; arriving at iteration 21 the two are indistinguishable, 10.7% [2, 24] against 10.3% [7, 13].

There is no evidence multi-objective BO is the more fragile of the two, and some that it is less. The point estimates for early error are half the scalar arm’s or less at every magnitude, a seventh at 0.05σ , a third at 0.25σ and half at 1σ and 5σ , and the intervals separate at 0.05σ and 0.25σ ; at 1σ they overlap slightly and at 5σ substantially, so this is a consistent direction rather than a clean separation at the top of the range. The onset ratio is correspondingly flatter: $3.0\text{--}5.5\times$ across magnitudes against the scalar arm’s $3.6\text{--}9.6\times$. A mechanism consistent with both observations, and with Appendix D.2, is that a hypervolume objective has no single incumbent to corrupt, so the *search* keeps finding good designs. The front the study would *report* is another matter: at 1σ from the first rating about half of it is truly dominated (5.2 of 10.9 designs), it misses most of the truly non-dominated designs evaluated, and its hypervolume shortfall exceeds that of the evaluated set in every cell.

Table 14: Hypervolume acquisitions, ranked over problems within each condition. Absolute loss is the fraction of the hypervolume floor gap lost under noise. The four are not distinguishable.

acquisition	mean rank	absolute loss
qNEHVI	2.40	20.2%
qLogEHVI	2.44	20.4%
qEHVI	2.56	20.7%
qLogNEHVI	2.60	20.8%

The choice among hypervolume acquisitions does not appear to matter. Mean ranks span 2.40 to 2.60 on a scale where 2.5 is a tie, absolute losses span 20.2% to 20.8%, and the Friedman test reaches $p < 0.05$ in 0 of 8 conditions (Table 14). Kendall’s W over condition-averaged losses is 0.056, against 0.52 for the scalar suite computed the same way, and each of the four acquisitions is the most robust on at least one of the six problems. Six problems at five seeds is far less power than twenty at ten, so this is “no detectable difference” rather than “no difference”. It does, though, point the same way as Section 8: the disagreement between design spaces about which acquisition is most robust is, if anything, sharper here.

What does not transfer. Equation 7 is defined for a scalar objective: it is the loss from one greedy pick among m candidates, and on a Pareto problem there is no single greedy pick whose loss it measures. Section 7’s result is therefore untested in this arm rather than confirmed or refuted by it. A hypervolume analogue would have to define the one-shot loss over hypervolume contributions and take a stand on how per-objective error propagates into them; that is a paper of its own, and we do not claim Table 6 extends until someone writes it.

F THREE DESIGN CHOICES THAT COULD HAVE PRODUCED THE RESULT

Every number above was measured at a fixed budget, against an unbounded continuous response, using standard acquisition functions. Each of those is a choice, and each could be doing the work. The three arms here ran GAUSSIAN error on seeds 7–11 with six acquisitions (LogEI, EI, PI, UCB, qUCB, qNEI) or, for the robust baselines, the two named; every reference row is the main sweep restricted to the same error process, acquisitions and seeds.

Table 15: Budget, with the late onset held at 40% of the run in each case and the reference row restricted to the arm’s own error process, acquisitions and seeds, so that budget is the only thing that changes. Percentages are the fraction of the achievable improvement destroyed.

budget	error present	0.05σ	0.25σ	1σ	5σ
$T = 25$	from it. 1	0.6	3.2	9.4	15.4
	from it. 11	0.3	1.3	4.1	8.0
	ratio	$2.2\times$	$2.4\times$	$2.3\times$	$1.9\times$
$T = 50$	from it. 1	0.6	3.8	11.7	21.3
	from it. 21	0.1	1.0	2.3	4.7
	ratio	$4.4\times$	$3.9\times$	$5.1\times$	$4.6\times$
$T = 100$	from it. 1	0.2	3.4	11.1	23.9
	from it. 41	0.1	0.4	0.9	2.2
	ratio	$1.4\times$	$9.2\times$	$11.7\times$	$10.8\times$

Budget, and the onset effect is not budget-free. Re-running at $T = 25$ and $T = 100$ with the onset held at the same fraction of the run gives Table 15. Early-error cost is roughly flat in budget — 9.4%, 11.7%, 11.1% at 1σ — but late-error cost falls steeply, from 4.1% at $T = 25$ to 2.3% at $T = 50$ to 0.9% at $T = 100$. The onset ratio therefore grows with the budget, from $2.3\times$ to $5.1\times$ to $11.7\times$. This is the clearest limitation on the headline: **the $5\times$ onset effect is a statement about a fifty-iteration study**. The direction is unsurprising, since more remaining iterations means more chance to recover, but the size is not: a fourfold change in budget moves the ratio by five. We looked for a rescaling that would let one budget speak for another and did not find one. Late-onset cost per remaining iteration falls by about $17\times$ across the three budgets, and late-onset cost times remaining iterations is non-monotone (0.62, 0.69, 0.57), so neither is the invariant. The early-error number is the transferable one; a study at a different budget has to measure its own onset effect.

Table 16: A bounded, discrete response. The instrument arm clips to $\pm 8\sigma$ and rounds to 0.55σ , the range and step implied by the three archival studies. Both rows: GAUSSIAN error, six acquisitions, five seeds.

response scale	error present	0.05σ	0.25σ	1σ	5σ
unbounded, continuous	from it. 1	0.6	3.8	11.7	21.3
	from it. 21	0.1	1.0	2.3	4.7
bounded, discrete	from it. 1	3.8	5.0	11.7	20.8
	from it. 21	1.3	1.4	2.5	4.6

A real rating scale, which cuts both ways. The synthetic objective is unbounded and continuous; a person returns a bounded, discrete rating. The three archival instruments have steps of 0.55 , 0.54 and 1.12σ and span sixteen to twenty σ , so the arm clips to $\pm 8\sigma$ and rounds to 0.55σ — the two five-item instruments, in the study’s own currency. Table 16 shows the two effects separately. At 0.05σ the discretised arm is six times worse (3.8% against 0.6%), because rounding to a 0.55σ grid is itself an error of about 0.16σ and swamps an injected error far smaller than the step; at 0.25σ it is still worse (5.0% against 3.8%); at 1σ the two are indistinguishable (11.7% against 11.7%); and at 5σ it is slightly better (20.8% against 21.3%), because clipping bounds how far one bad rating can move the incumbent. A rating scale is a noise floor at the bottom of the range and, at the top, a marginal safety rail.

Table 17: Robust baselines. qKG is the knowledge gradient; replication spends every second evaluation re-asking about the incumbent. Compared at a fixed number of evaluations, so replication buys averaging with half the designs. The standard row is the ten model-based acquisitions of the main sweep on the same error process and seeds.

acquisitions	error present	0.05σ	0.25σ	1σ	5σ
ten standard	from it. 1	1.1	5.6	13.9	21.5
	from it. 21	0.3	1.3	2.7	4.5
qKG and replication	from it. 1	0.3	2.2	9.2	16.2
	from it. 21	0.2	0.1	2.2	4.8

Methods built for noise do not recover the cost. All ten model-based acquisitions above are ordinary ones run under noise. Two baselines built for it: the knowledge gradient, which values the improvement in the posterior *maximum* rather than in any observed value, and spending every second trial re-asking about the incumbent. Scored the way the rest of this appendix is, against each method’s own clean run, they lose less: 9.2% of the achievable improvement at 1σ from the first rating against the standard acquisitions’ 13.9%, and 16.2% against 21.5% at 5σ (Table 17). That comparison flatters them. Replication re-rates the incumbent in its clean run too, which carries no information, and qKG’s clean run is itself worse than the standard acquisitions’, so part of the smaller excess is a worse reference. Scored against the standard process’s clean run (Appendix Q), replication recovers none of the cost (−61% on the trajectory, −5% on the deployed design) and qKG recovers some of it at the early onset (+12% to +43% at 0.25σ and above) but more than loses it at the late onset and in the design it recommends (−32%). What remains is the information the error destroys, which is what Equation 7 measures.

G HELD-OUT SEEDS: A VOID REPLICATION AND ITS SENSITIVITY ANALYSIS

Everything above was selected and estimated on the same ten seeds. Before running any fresh ones we wrote down four claims, the statistics that would test them, the thresholds each had to clear, and four control conditions that would void the run, then ran ten fresh seeds under GAUSSIAN error only, as preregistered.

Table 18: Preregistered quantities on the screening seeds and on ten fresh ones. Both columns use GAUSSIAN error only and exclude Rosenbrock, so the screening column differs slightly from the pooled twenty-landscape values of Sections 4–8. The replication line reports the *sensitivity* analysis described in the text, not the preregistered run, which is void.

preregistered quantity	screening	fresh seeds
H1 $\beta(\text{frag})$	+0.65	+0.69
H1 $\beta(\log_{10} \sigma_e)$	−0.06	−0.07
H2 onset ratio	5.44	5.01
H3 robust mean rank	4.24	4.11
H3 fragile mean rank	7.26	6.98
H4 dose-response monotone	yes	yes
replicates — H1: yes; H2: yes; H3: yes; H4: yes		

The preregistered run is void, by its own rule. One control failed: Rosenbrock’s headroom on the fresh seeds is 0.083, below the 0.10 the preregistration fixed in advance. The document says that a control failure makes the run void rather than negative, and an exclusion applied after seeing the data cannot undo that. We report it as void.

We also report what it showed, labelled as the post-hoc sensitivity analysis it is. Dropping Rosenbrock from both arms, all four claims clear their thresholds: $\beta(\text{frag})$ moves from +0.65 to +0.69 while $\beta(\log_{10} \sigma_e)$ moves from −0.06 to −0.07 with an interval containing zero; the onset ratio is 5.01 against 5.44; the robust and fragile acquisition families sit at mean ranks 4.11 and 6.98 against 4.24 and 7.26; and the dose-response is monotone in both (Table 18). Two deviations from the doc-

ument should be named. The script tested the family contrast over the eight conditions rather than the twenty benchmarks as preregistered; run as preregistered it gives $p = 3.6 \times 10^{-5}$ over the twenty benchmarks and 7.2×10^{-5} without Rosenbrock. And the null on the error magnitude holds under the preregistered additive specification only: with the magnitude-by-opt_z interaction of Section 7 the magnitude is significant again, so that sub-claim confirms a specification, not a fact about the error.

One argument is worth making explicitly, because it decides how much the void should worry a reader. The headroom control exists to stop a benchmark with nothing to lose from contributing a null that looks like evidence of robustness. Five of the sub-claims are positive results, and including a low-headroom benchmark makes those harder to obtain, not easier. The two null-type sub-claims, that the magnitude’s interval contains zero and that Kendall’s W is below 0.40, run the other way, and there the sensitivity estimates are within 0.01 of the void run’s (interval $[-0.13, 0.09]$ against $[-0.13, 0.08]$; W 0.272 against 0.276). That is a reason to read the sensitivity analysis as informative; it is not a reason to call the run confirmed, and we do not.

H THE LANDSCAPES AND THE DESCRIPTOR FIT

Table 19 lists the twenty landscapes with their measured geometry, and Table 20 places the three human studies of Section 4 on the same axis. Section 7 reports the descriptor model only as a comparison for frag(σ_e); Table 21 gives it in full, with the variance inflation factors that limit what can be read off it.

Table 19: The twenty landscapes and their measured geometry. opt_z is how many landscape standard deviations the optimum stands above a random design; sparsity is the fraction of the box within 10% of the optimum; ruggedness is 1 minus the lag-1 autocorrelation along random walks; tail is $\sigma/(1.4826 \text{ MAD})$; frag(1) is the one-shot selection loss at an error of one landscape standard deviation. Names are the suite’s keys; levy_10 has eleven inputs.

benchmark	d	opt _z	sparsity	rugged.	skew	tail	frag(1)
powell	4	0.76	7.2e-01	0.13	−2.33	2.52	0.18
rosenbrock	4	1.03	4.9e-01	0.13	−1.06	1.13	0.18
branin	2	1.25	2.4e-01	0.13	−1.09	1.14	0.30
steering_law	4	1.29	7.9e-01	0.12	−3.33	1.44	0.43
power_law_practice	4	1.72	1.4e-01	0.13	−2.02	1.72	0.63
weber_fechner	4	2.15	5.5e-02	0.15	−0.71	1.01	0.64
levy_10	11	2.62	1.3e-02	0.20	−0.54	1.00	0.80
griewank	6	2.75	1.0e-02	0.12	−0.26	0.98	0.85
moving_peaks	4	2.96	1.8e-03	0.17	−0.53	1.04	0.64
hartmann_3	3	3.05	1.6e-02	0.14	+1.14	1.49	0.50
stevens	4	3.06	3.0e-03	0.12	−0.17	0.97	0.75
eggholder	2	3.21	6.6e-03	0.42	+0.04	1.00	0.68
yerkes_dodson	4	3.54	1.9e-03	0.12	+0.27	0.98	0.73
rastrigin	4	3.64	5.8e-04	0.58	−0.08	0.98	0.69
schwefel	4	4.33	1.2e-04	0.25	+0.00	0.99	0.69
hicks_law	4	5.40	<1.5e-05	0.12	+0.40	0.97	1.00
hartmann_6	6	7.97	9.2e-05	0.14	+2.67	2.74	0.53
michalewicz	5	8.05	<1.5e-05	0.36	+1.02	0.92	0.80
ackley	4	20.10	<1.5e-05	0.80	+3.26	1.67	0.18
shekel	4	56.77	<1.5e-05	0.13	+3.72	1.63	0.00

opt_z carries a VIF of 9.2 and regresses on the other descriptors with $R^2 = 0.89$, so its coefficient and those of sparsity and skew (excluded here, VIF 5.8 and 6.3) are not separately identified. This is why Appendix D.3 builds landscapes that vary these properties independently rather than trying to disentangle them by regression. Dimension and ruggedness are null here; the fixed-opt_z ladder of Appendix D.3 shows dimension does raise the cost, so that null is a limit of the observational design.

Table 20: Three published human studies placed on the synthetic arm’s axis. σ_f is the spread of the study’s fitted objective over its design box; the last column is the measured within-rater noise in those units.

study	σ_f	opt_z	rating noise	noise / σ_f
ehmi	0.365	2.37	0.269	0.74
opticarvis	0.374	13.36	1.254	3.35
provoice	0.298	4.00	0.304	1.02

Table 21: Post-onset excess regret on landscape descriptors and the error magnitude, predictors z -scored, cluster-bootstrapped over landscapes with Benjamini–Hochberg correction. VIF is computed on the design matrix.

term	β	95% CI	p_{FDR}	VIF
$\log \sigma_e$	+0.37	[+0.15, +0.66]	0.001**	1.0
$\log \text{opt}_z$	+0.64	[+0.24, +0.74]	0.001**	9.2
$\log \text{tail}$	+0.20	[+0.03, +0.33]	0.008**	1.2
rugged.	−0.09	[−0.38, +0.12]	0.519	1.3
d	−0.03	[−0.26, +0.12]	0.768	1.0
$R^2 = 0.526$, 640 cells, 20 landscape clusters				

I INPUT ERROR: THE DESIGN THAT SHIPS

Under an unnoticed slip the optimizer’s record holds the proposal x , not the point x' it evaluated, and the design a practitioner deploys is the recorded one. Table 23 scores both at the end of the run: the best design the person actually experienced, and the design the log recommends, scored where the log says it is.

Table 22: Where a slip’s cost comes from, at both onsets. *floor*: the model-free arms, which evaluate displaced points but learn nothing; *logged*: a learner whose record holds the point actually evaluated; *unnoticed*: the same learner with the record holding the proposal; *mislabelling*: the paired difference of the last two, on shared seeds and slip draws. The early-onset rows also appear in Table 5.

SLIP	floor	logged	unnoticed	mislabelling
<i>error from it. 1</i>				
1%	0.3 [0, 1]	0.7 [0, 2]	1.5 [0, 3]	0.8 [0, 2]
5%	0.2 [−1, 1]	2.8 [2, 4]	5.4 [3, 8]	2.7 [1, 4]
15%	1.2 [−1, 3]	7.1 [5, 10]	11.5 [8, 15]	4.4 [2, 6]
40%	6.4 [0, 13]	19.6 [12, 28]	23.2 [16, 31]	3.6 [2, 6]
<i>error from it. 21</i>				
1%	0.0 [0, 0]	0.2 [0, 1]	1.1 [0, 2]	0.9 [0, 1]
5%	−0.4 [−1, 0]	2.0 [1, 3]	3.0 [2, 5]	1.0 [0, 2]
15%	−0.4 [−1, 1]	4.5 [2, 7]	5.0 [3, 8]	0.5 [0, 1]
40%	−0.6 [−2, 1]	6.3 [3, 10]	6.4 [4, 10]	0.1 [0, 0]

Table 23: Final-iteration excess simple regret under an unnoticed slip, as a percentage of the achievable improvement, over the model-based acquisitions. *Evaluated*: the best true value among the points actually touched. *Deployed*: the true value at the recorded location of the best-rated trial. Brackets are 95% cluster-bootstrap intervals over landscapes.

SLIP	from it. 1		from it. 21	
	evaluated	deployed	evaluated	deployed
1%	1.7 [0, 3]	2.7 [1, 5]	1.5 [1, 3]	2.4 [1, 4]
5%	6.5 [4, 10]	16.4 [9, 24]	3.9 [2, 6]	9.6 [5, 15]
15%	13.4 [8, 20]	34.3 [24, 45]	6.9 [3, 10]	13.4 [8, 19]
40%	24.9 [16, 34]	65.8 [53, 78]	9.0 [5, 14]	13.3 [8, 19]

J EXTRA TRIALS, BY ARM

Table 24 repeats the count of Section 4 for every arm at the early onset, against a clean ten-trial study within the fifty-trial budget, with the same one-percent tolerance. Fewer than half the runs are censored in every cell, so every median is exact; the strict count with no tolerance is in `extra_runs.csv` beside each arm. The recorded slip costs fewer trials than the unnoticed one at every magnitude above 1%, and a misclick almost none below 40% of the trials.

Table 24: Median extra trials to match a clean ten-trial study within 1% of the achievable improvement, error from the first trial, with the share of runs that never do within 50 trials. Response-error rows are the main sweep over all ten model-based acquisitions and ten seeds; input-error rows are six acquisitions and five seeds.

error	0.05 σ	0.25 σ	1 σ	5 σ
GAUSSIAN	0 (1%)	1 (8%)	9 (26%)	26 (44%)
BIAS	0 (1%)	2 (9%)	8 (27%)	28 (45%)
DRIFT	0 (1%)	2 (9%)	8 (28%)	36 (49%)
AR(1)	0 (1%)	1 (4%)	3 (22%)	23 (42%)
error	1%	5%	15%	40%
SLIP	0 (3%)	3 (7%)	8 (15%)	24 (40%)
SLIP, logged	0 (2%)	2 (5%)	4 (10%)	14 (25%)
MISCLICK	0 (1%)	0 (3%)	0 (9%)	6 (23%)

K RE-SCORING THE DESIGN A STUDY SHIPS

The deployed regret everywhere else in this paper uses one ship rule: the design with the single best observed rating. To separate search from selection (Section 5) we re-score what the same trials would have shipped under other rules, adding no trial. The loop’s own surrogate — BoTorch’s default single-task GP, normalised over the benchmark’s box and standardised — is refitted on every logged observation, and each rule picks among the *visited* designs: the best single rating (the standard), the best mean rating, the posterior-mean maximiser, the maximiser of the posterior mean less one or two latent SDs, and, as the ceiling, the visited design with the best true value. The arm’s own logged rule must reproduce the logged final regret to 10^{-9} in every run or the analysis stops; over the 66,000 runs it did, with no failed fits.

Scored as a share of the deployed cost of GAUSSIAN error, the cautious rule (posterior mean less two latent SDs) recovers +5.8% [2, 10] at 1 σ from the first rating, +12.6% [4, 23] at 1 σ from trial 21, +9.7% [5, 15] and +18.5% [11, 26] at 5 σ . It pays a fixed price of 0.015 of the achievable improvement whether or not error is present, so below 0.25 σ it costs 80–150% more than the error does and should not be used without an estimate of the rater’s noise. Under BIAS it survives only at 1 σ (+5.5% [1, 9]); under DRIFT and AR1 nothing survives correction.

L END-OF-STUDY PROCEDURES

Both procedures keep T fixed and spend the final trials differently, so each is replayed exactly from an existing log: truncate at the trial the procedure starts, refit the loop’s GP on that prefix, and simulate only the new trials. The premise is checked against the re-rating arm, whose earlier trials must equal the standard run’s.

Tournament(k). Search trials $1..T - k$; the last k are one sitting in which each of k candidates is shown once. Candidates are the top k visited designs by the cautious ship rule, or by observed rating; the winner is the best look, or the posterior mean after conditioning on the looks. Error shared by every trial of a sitting cancels under comparison, so only the idiosyncratic part enters: σ for GAUSSIAN, BIAS and DRIFT, the innovation for AR1, and we report the sitting at both that SD and half of it.

Confirmation(k). Search trials $1..T - 2k$; the last $2k$ rate the chosen design and the best initial design k times each in counterbalanced order, and the chosen design ships only if a one-sided paired t -test at $\alpha = 0.05$ prefers it. This changes little about the design that ships (it recovers 3.2%–3.4% of the cost) and a great deal about what the study may claim. The standard process claims an improvement in 98.3% of runs and that claim is false in 15.7% of them, rising to 42.7% at 5σ . Four confirmation trials take the false-claim rate to 0.5% overall and 1.0% at 5σ , at the cost of detecting only 50.7% of the real improvements; eight trials give 0.46% and 65.7%.

M DERIVING THE SPLIT FROM THE RUN IN HAND

Write μ_i, σ_i for the loop GP’s posterior latent mean and SD at the distinct designs visited by the decision point, ordered by the cautious ship rule, and s for the idiosyncratic rating SD of a sitting. A k -tournament shows the top k candidates once each, so with $f_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ and looks $y_i = f_i + \mathcal{N}(0, s^2)$ the identification term is $U(k) = \mathbb{E}[f_{\arg \max_i y_i}]$, evaluated by Monte Carlo. It is not monotone in k : a larger candidate set is more likely to contain the best visited design and more likely to let a worse one win on a lucky look. The search term prices the forgone trials by the run’s own gain in best rating per trial over the last ten, floored at zero. Every policy — ship by posterior mean, ship by the cautious rule, or a tournament of k — is scored under the single posterior at $n_0 = T - \max k$, the last trial at which all of them are still open; scoring each k under its own $T - k$ posterior looks more faithful and is not, because two GPs fitted on different numbers of points put their maxima on different scales, and the comparison then picks up that difference rather than the difference between policies. The sitting’s SD is read from the surrogate’s fitted noise, which is the noise of the *standardised* targets, and multiplied by the outcome transform’s scale so that it and the posterior sit on one axis; without that the rule read a sitting as up to 5.6 times more precise than it is. The rule reads only ratings and the surrogate; the true objective enters only when the choice is scored, against the realised regret of that (run, k) in the replay.

N BUDGET-NEUTRAL ARMS

Every arm below keeps the number of human trials equal to the standard process or lowers it, and is scored against the standard process by the estimand of Appendix Q: twenty landscapes, seeds 7–11, both onsets where the arm has them. Table 25 gives the share of the standard process’s cost of error that each recovers on the design the study deploys.

The pattern is not that everything fails. It is that what fails is the acquisition, and what works is an explicit model of how the rating was produced. Clipping the response to the range actually observed recovers 22–35% of the cost of gross faults at a price of exactly zero; a relevance-pursuit likelihood recovers 26–38% when 15% of trials carry a gross fault and loses when only 5% do; fitting per-rater offsets recovers 32% [23, 40] when raters alternate and nothing when they hand over in blocks, because a block offset is confounded with the search’s own time trend while an interleaved one is identifiable; re-anchoring a saturating cap recovers 25% [17, 32]; and removing the factor shared across objectives recovers 25% [7, 50] of the deployed cost of halo error. A saturating rating scale is itself among the most expensive faults we measured, costing 185% [132, 258] more than the same rating noise on a scale that does not saturate.

Table 25: Budget-neutral arms: share of the standard process’s cost of error recovered on the deployed design, with 95% landscape-bootstrap intervals. Negative means the change does worse under error than the standard process. Above the rule, changes to the search; below it, changes to the model of how the rating was produced.

Arm	Recovered
Augmented expected improvement	+2% [−6, +11]
Slip-aware acquisition, 0.05 slip	−4% [−15, +4]
Effort front-loaded to the first ten trials	−7% [−20, +4]
Effort front-loaded to the first twenty	−8% [−31, +5]
Effort on the last ten trials	−12% [−24, −2]
Minimum-distance proposals	−13% [−23, −6]
Slip-aware acquisition, 0.15 slip	−18% [−34, −9]
Effort split between first and last trials	−21% [−33, −10]
Lower-confidence-bound incumbent	−27% [−42, −11]
Slip-aware acquisition, 0.4 slip	−35% [−54, −16]
Thompson sampling	−58% [−81, −40]
A lost rating imputed low rather than dropped	−979%
Relevance pursuit, 15% of trials spiking at 20 SD	+38% [+24, +51]
Response clipped to range, 5% spiking at 20 SD	+35% [+23, +46]
Per-rater offsets, round-robin handover	+32% [+23, +40]
Relevance pursuit, 15% spiking at 5 SD	+26% [+7, +42]
Re-anchored rating cap	+25% [+17, +32]
Halo backfit across objectives	+25% [+7, +50]
Response clipped to range, 5% spiking at 5 SD	+22% [+10, +35]
Response clipped to range, 15% spiking at 20 SD	+22% [+4, +38]
Response clipped to range, 15% spiking at 5 SD	+6% [−7, +18]
Per-rater offsets, block handover	+5% [−5, +14]
Relevance pursuit, 5% spiking at 20 SD	−30% [−119, +12]

O COMPARISONS INSTEAD OF RATINGS: WHAT THE INVARIANCE DOES AND DOES NOT BUY

Preference elicitation is the usual response to an unreliable rater (Chu & Ghahramani, 2005; González et al., 2017), on the argument that a comparison is invariant to how the rater uses the scale. The argument is exact and its reach is narrow, and both halves are measurable here. A fault is *shared* when it acts on every design of one sitting through the same monotone map g ; then for strictly increasing g , $g(f(a)) > g(f(b)) \Leftrightarrow f(a) > f(b)$, and the comparison is exactly what it would have been with no fault. A fault drawn afresh for each judgement cancels nothing, and a comparison inherits the noise of both designs.

We ran two loops at equal human cost on the twenty landscapes, three seeds, $T = 20$ judgements: the standard rating loop, and a comparison loop with a Laplace-approximate PAIRWISEGP (Balandat et al., 2020) on which each trial proposes one design and asks for a comparison with the incumbent, shipping the posterior-mean maximiser. Ten of the 960 fits fell back to prior hyperparameters and are counted, not dropped. Each loop is scored against its own identically seeded clean twin.

The invariance holds exactly and buys exactly what it should. Under DRIFT the rating loop’s shipped design differs from its clean twin in 41 of 60 runs, at an excess of 0.040 of the achievable improvement; the comparison loop’s differs in *none*, at an excess of 0.000 — not approximately, in no run. Under a saturating scale, a map that is monotone but not strictly so, the comparison buys nothing (−0.3% [−30, +27]): two designs above the cap are indistinguishable, and near the optimum that is exactly the pair the study must tell apart. Under GAUSSIAN noise, drawn afresh per judgement, the comparison is *worse* by 26.5% [−79, +15], the direction the $\sqrt{2}$ argument predicts. A constant BIAS present from the first trial moves neither loop, in no run: a rating loop that standardises its outputs and ships an argmax is already shift-equivariant, so that arm tests the rating loop’s invariance rather than the comparison’s, and the cost of BIAS elsewhere in this paper comes from its arriving *part-way through*.

So changing the elicitation is not a general remedy. It removes, exactly, that part of the fault which is a strictly monotone reparametrisation of the scale — drift, a rater’s personal offset, a handover between raters — and nothing of the inconsistency, the saturation or the gross faults.

P IS THE CUSUM THE RIGHT DETECTOR?

The freeze rule of Section 9 detects a mid-study change with a one-sided CUSUM on the squared one-step-ahead standardised residual of the loop’s own surrogate. A CUSUM is a choice, not a derivation, so we compare it with two alternatives on exactly the same 13,200 residual streams (6,600 tuning, 6,600 held out; 6,400 carrying a real mid-study change). Since the residual is standard normal while the rater behaves and a rater who becomes noisier inflates its variance, every detector tests $\mathcal{N}(0, 1)$ against $\mathcal{N}(0, \sigma^2)$, $\sigma > 1$: the CUSUM; the generalised likelihood ratio maximised over the unknown change point, whose statistic $\max_{\tau} \frac{n}{2} (\hat{\sigma}_{\tau}^2 - 1 - \log \hat{\sigma}_{\tau}^2)$ is the classical answer to that question and what a CUSUM approximates; and Bayesian online change-point detection (Adams & MacKay, 2007) with a constant hazard and a normal-inverse-gamma model of the post-change variance, alarming when the posterior probability that the current run is short exceeds a threshold. Each detector’s threshold is tuned on the tuning seeds to a common false-alarm budget — an alarm is false when the rater never changed, or when it fires at or before a late onset — and scored on the held-out seeds.

The detector is not the bottleneck. At a 10% budget the GLR detects 45.5% of changes and the CUSUM 45.4%; at 5%, 36.8% against 36.9%; at 20%, 56.0% against 56.4%. The CUSUM matches the classical optimum for this alternative to within half a percentage point at every budget, so there is nothing to be gained by replacing it. Bayesian online detection is worse on detection rate at every budget (23.0%, 32.2%, 45.1%) but alarms with a median delay of 2 trials against the CUSUM’s 6–9. Since the freeze ceiling falls from 82% to 54% after two trials of delay (Section 9), that trade is not obviously a bad one, but it is a choice of operating point rather than a better method. What limits the freeze rule is the signal — a rater who becomes somewhat noisier midway simply does not leave much of one — not the statistic computed from it.

Q PROCESS ADAPTATIONS

The measurements above say what error costs a fixed process. The practical question is whether the process can be changed to cost less, and the scoring has to change with it. Rating a design twice, or re-rating designs before deploying, also changes the clean run — a repeated rating of an exact objective carries no information — so a process change scored by its excess over its own clean run is measured against a handicapped reference. We score each change against the *standard* process instead. Per paired cell (landscape, acquisition, magnitude, onset, seed), with regret as a fraction of the achievable improvement,

$$\text{recovered} = \frac{R_{\text{noisy}}^{\text{std}} - R_{\text{noisy}}^{\text{adapt}}}{R_{\text{noisy}}^{\text{std}} - R_{\text{clean}}^{\text{std}}}, \quad \text{price} = R_{\text{clean}}^{\text{adapt}} - R_{\text{clean}}^{\text{std}},$$

aggregated as ratios of landscape means with landscape-bootstrap intervals, on two responses: the post-onset trajectory regret and the regret of the design the experimenter would deploy. qKG and replication are different acquisitions, so there the standard process is the mean over the ten standard acquisitions. Every arm runs twenty landscapes, seeds 7–11, both onsets and the error process named in Table 26, with LogEI and qNEI unless stated; the incumbent and known-noise rows re-score the arms of Appendix D.1 and D.2 with their own acquisitions.

Two of the paper’s own ablations, re-scored. The observed-max incumbent and the known-noise GP change the clean run too, so their shares in Appendix D.1 and D.2 are also excess over their own clean runs. Against the standard process the conclusions stand, with one addition. The incumbent recovers 19% [10, 27] of the trajectory cost at a price of 0.1 points without error, most of it from the first rating (+10% to +67% across magnitudes), and cuts the extra trials needed to match a clean standard ten-trial study at 1σ from 8 (21% never) to 6 (13%), and at 5σ from 36 (48%) to 21 (39%). The design it deploys is worse, though, by 10% of the cost [5, 15], mostly at the late onset, for a reason we have not isolated. Giving the GP the true noise variance recovers nothing on either response (−3%).

Table 26: Share of the standard process’s cost of error that each process change recovers, pooled over magnitudes and onsets, with 95% landscape-bootstrap intervals (a star when the interval excludes zero), and the price the change pays without error in percentage points of the achievable improvement. Negative recovery means the adapted process does worse under error than the standard one. The fitted-oracle datasets have no achievable-improvement scale.

adaptation	recovered (%)		price without error	
	trajectory	deployed	trajectory	deployed
rating error (GAUSSIAN)				
observed-max incumbent	+19* [10, 27]	−10* [−15, −5]	+0.1	+0.2
GP given the true noise variance	−3 [−16, 8]	−3 [−11, 5]	+0.4	−0.0
first ten proposals rated twice	−88* [−113, −67]	−14* [−21, −8]	+5.1	+1.6
+ observed-max incumbent (LogEI)	−93* [−127, −58]	−44* [−57, −32]	+4.7	+1.1
last six trials re-rate the top three	−1* [−2, −1]	+4* [0.04, 9]	+0.1	+0.8
knowledge gradient	−4 [−23, 14]	−32* [−42, −24]	+2.7	+3.6
re-rate the incumbent every second trial	−61* [−88, −33]	−5 [−12, 3]	+5.4	+4.3
unnoticed SLIP				
noisy-input GP	−28* [−43, −13]	−19* [−31, −9]	−0.1	−1.5
last six trials re-rate the top three	0 [−0.3, 0.2]	−6 [−15, 1]	+0.1	+0.8
MISCLICK				
Student- <i>t</i> surrogate	−2 [−38, 31]	−13 [−50, 16]	+0.2	+1.5
fitted oracles, rating error				
first ten proposals rated twice	−161* [−3504, −49]	−8 [−17, 6]	−	−

Rating early designs twice makes things worse. The onset effect says the first trials are where error is expensive, so replicating them is the natural fix. It is not one: rating each of the first ten proposals twice recovers −88% of the trajectory cost and −14% of the deployed design’s (Table 26), and it needs more extra trials to match a clean standard ten-trial study, a median of 8 at 1σ against 6 for the standard process. In a fifty-trial study the ten designs it gives up cost more than averaging two ratings, which only divides the error variance by two, buys; the GP already pools neighbouring observations. Adding the observed-max incumbent of Appendix D.2 does not rescue it (−44% on the deployed design), and the same replication on the three fitted-oracle datasets points the same way (−8% on the deployed design, interval [−17, 6]). Re-rating the incumbent every second trial fails for the same reason (−61% on the trajectory). qKG does not replicate: it helps with error from the first trial and hurts with late error and on the deployed design, for a reason we have not tested (Appendix F).

Re-evaluating before deploying helps, a little. Spending the last six trials re-rating the three designs with the best mean rating, and deploying by mean, costs almost nothing in the trajectory (−1%) and recovers 4% of the deployed design’s cost, [0.04, 9], reaching 18% [5, 29] when large error arrives late. Under an unnoticed slip it recovers nothing (−6%, [−15, 1]): the re-ratings land on displaced points too, so averaging them does not correct the record.

A noisy-input GP helps small early slips and hurts large late ones. Inflating each point’s observation variance by the squared gradient of the posterior mean times the slip variance recovers 76% [27, 245] of the cost of a 1% slip from the first trial and 17% of a 5% slip, but at the late onset it recovers −129% and −205% of the cost of 15% and 40% slips. A likely reason, which we have not tested, is that by then the search sits near a steep peak whose flanks have the largest gradients, so the observations that locate the peak are discounted most. Pooled it recovers −28% of the trajectory cost.

A heavy-tailed likelihood does not absorb misclicks. A misclick records the rating of a random design against the proposed one, an outlier rather than noise, so a Student-*t* likelihood should discount it. A variational Student-*t* GP in place of the standard one recovers −2% [−38, 31] of the trajectory cost and −13% [−50, 16] of the deployed design’s, at a price of 0.2 and 1.5 points without error. Where misclicks cost most, on 40% of trials from the first, it recovers −1% [−10, 6] of the trajectory cost, and the median run still needs 8 extra trials to match a clean standard ten-trial study, as it does without it (24% never, against 28%). Misclicks cost little in the first place, so there is little for any model to recover.

R THE ONE-SHOT LOSS FROM A PILOT

Equation 7 is defined on the exact objective at 256 quasi-random designs, which a practitioner does not have. For each landscape and each of the ten seeds we fit the study’s own GP to the first k trials of a clean LogEI run and evaluate Equation 7 on its posterior mean, with the same 256 designs and the same 4,000 error draws as the exact value (Table 27).

Table 27: Spearman correlations across the twenty landscapes of frag computed from the GP posterior mean after k clean trials (averaged over ten seeds) with the exact frag and with the measured cost at the early onset. Pooled rows use all eighty landscape-by-magnitude cells.

frag computed from	with exact frag	with measured cost		
	pooled	pooled	1σ	5σ
first 10 trials	0.94	0.85	0.32	0.61
first 20 trials	0.96	0.85	0.20	0.66
first 50 trials	0.96	0.85	0.15	0.63
exact objective	1.00	0.84	0.42	0.45

Ten trials are enough. Pooled over magnitudes, frag from the first ten trials correlates 0.94 with the exact value and 0.85 with the measured cost, against 0.84 for the exact frag itself; twenty and fifty trials add almost nothing. Within a single magnitude the ranking of landscapes is weak for the exact frag too (0.08 to 0.45), so the predictive power is mostly across magnitudes; at 5σ the pilot underestimates frag by about a third on average (a bias of -0.78) but ranks landscapes by cost better than the exact version (0.61 against 0.45).

S DOES NOISE EVER HELP?

A noisy observation can in principle help, by pushing an over-exploiting optimizer out of a local basin. At one seed per cell, 25 of the 640 landscape-by-condition cells had negative mean excess regret with $d_z < -0.2$, noise apparently improving the outcome, which is the kind of finding that invites a mechanism.

At ten seeds it is one cell: michalewicz under AR(1) error at 0.05σ from iteration 21, mean excess -0.136 , $d_z = -0.21$ over 100 paired runs. One cell in 640 is fewer than the fifteen or so that a null family of 640 tests would flag at this threshold. We report it as a null and note the sampling-variability lesson: the twenty-four cells that disappeared between one seed and ten were all in the direction that would have made the better story.